DATA WAREHOUSING ASSIGNMENT

Movie Rental Data Warehouse Design

Prepared by: Anas Saleh & Nada Ishtaya

Course: Data Warehousing / Data Architecture

Course Instructor: DR. Moamin Abughazala

Topic: From OLTP Schema to Dimensional Model and ETL Design

1. Introduction

This report proposes a high-level data warehouse design for a movie rental business. The source system is an OLTP database (Sakila) that supports daily operational activities including customer registration, film management, inventory tracking, rental recording, payment processing, staff management, and store and location management.

Although the OLTP system is well-suited for transactional processing, it is not optimized for analytical reporting. The data is highly normalized, spread across many related tables, and designed to minimize redundancy and support fast inserts, updates, and deletes. Business managers, however, need to answer analytical questions about rental trends, revenue performance, customer behavior, and film popularity. For this purpose, a separate data warehouse is required.

The proposed data warehouse reorganizes operational data into a dimensional model following Ralph Kimball's star schema methodology. This model simplifies the schema, reduces the number of joins required for reporting, and structures data around business processes that analysts can easily query.

The following table summarizes the key differences between the OLTP system and the proposed data warehouse:

Characteristic	OLTP System	Data Warehouse
Primary Purpose	Daily transaction processing	Analytical reporting and decision-making
Schema Design	Highly normalized (3NF)	Dimensional model (Star Schema)
Data Content	Current operational data only	Historical and summarized data over time
Optimized For	Fast insert, update, delete	Fast read and aggregation queries
Query Type	Simple transactional queries	Complex analytical queries with aggregations
Users	Staff and operational systems	Managers, analysts, business decision-makers
Data Volume	Moderate, current records	Large, historical records over years

2. Business Questions

Before designing the data warehouse, it is essential to identify the analytical questions the business needs to answer. The following business questions were derived by studying the OLTP schema and understanding the key business processes: rental activity, payment processing, inventory management, customer behavior, and store performance.

2.1 Film and Category Analysis

1. Which films are rented most frequently?
2. Which films generate the highest revenue?
3. Which film categories are most popular among customers?
4. What is the average rental duration for different films or categories?
5. Which films are returned late most often?

2.2 Store and Revenue Analysis

6. Which stores generate the highest number of rentals?
7. Which stores generate the highest revenue?
8. How does store performance differ by location?
9. How does revenue change by month, quarter, or year?

2.3 Customer Behavior Analysis

10. Which customers rent the most films?
11. Which customers generate the highest revenue?
12. What are the most active customer locations?
13. Which cities or countries generate the highest customer activity?

2.4 Time-Based Trend Analysis

14. How does rental activity change over time?
15. Which staff members process the highest number of rentals or payments?

These questions are important because they provide business managers with insights to make data-driven decisions: identifying popular films helps with procurement, revenue analysis supports financial planning, understanding customer behavior allows for targeted promotions, and trend analysis reveals seasonal patterns and operational bottlenecks.

3. Dimensional Model Design

3.1 Selected Business Processes

After analyzing the OLTP schema, three main business processes were identified as candidates for fact tables in the data warehouse:

- Film rental transactions: every time a customer rents a film from a store
- Payment transactions: every time a customer makes a payment for a rental
- Inventory availability: the availability status of each physical film copy in each store

3.2 Fact Tables

Fact Table	Business Process	Grain	Main Measures
Fact_Rental	Rental transaction	One row per rental transaction	rental_count, rental_duration_days, late_return_flag, late_days
Fact_Payment	Payment transaction	One row per payment transaction	payment_amount, payment_count
Fact_Inventory_Snapshot	Inventory availability	One row per inventory item per store per snapshot date	inventory_count, available_flag

3.2.1 Fact_Rental

Fact_Rental stores one record for every film rental transaction. This is the central fact table of the data warehouse and connects rental activity to all relevant dimensions. It supports analysis of rental frequency, late returns, rental duration, and customer and store behavior over time.

Column	Type	Description
rental_fact_key	INT (PK)	Surrogate primary key
rental_id	INT	Operational rental ID from OLTP
rental_date_key	INT (FK)	Foreign key to Dim_Date (rental date)
return_date_key	INT (FK)	Foreign key to Dim_Date (return date)
customer_key	INT (FK)	Foreign key to Dim_Customer
film_key	INT (FK)	Foreign key to Dim_Film
store_key	INT (FK)	Foreign key to Dim_Store

Column	Type	Description
staff_key	INT (FK)	Foreign key to Dim_Staff
inventory_key	INT (FK)	Foreign key to Dim_Inventory
rental_count	INT	Always 1 per row; used for aggregation
rental_duration_days	INT	Actual number of days the film was rented
late_return_flag	BOOLEAN	1 if returned late, 0 otherwise
late_days	INT	Number of days past the allowed rental period

3.2.2 Fact_Payment

Fact_Payment stores one record per payment transaction. It supports revenue analysis by customer, staff, store, and time period. Each payment is linked to a rental, allowing joining of rental and payment data when needed.

Column	Type	Description
payment_fact_key	INT (PK)	Surrogate primary key
payment_id	INT	Operational payment ID from OLTP
rental_id	INT	Links payment back to rental
payment_date_key	INT (FK)	Foreign key to Dim_Date
customer_key	INT (FK)	Foreign key to Dim_Customer
staff_key	INT (FK)	Foreign key to Dim_Staff
store_key	INT (FK)	Foreign key to Dim_Store
payment_amount	DECIMAL(10,2)	Dollar amount of the payment
payment_count	INT	Always 1 per row; used for aggregation

3.2.3 Fact_Inventory_Snapshot

Fact_Inventory_Snapshot stores the availability status of each film copy in each store at a given point in time. It is a periodic snapshot fact table that supports inventory analysis, showing which films are available or currently rented out.

Column	Type	Description
inventory_snapshot_key	INT (PK)	Surrogate primary key
date_key	INT (FK)	Foreign key to Dim_Date (snapshot date)
inventory_key	INT (FK)	Foreign key to Dim_Inventory
film_key	INT (FK)	Foreign key to Dim_Film

Column	Type	Description
store_key	INT (FK)	Foreign key to Dim_Store
inventory_count	INT	Always 1 per row
available_flag	BOOLEAN	1 if available, 0 if currently rented

3.3 Dimension Tables

Dimension	Main Attributes	Source OLTP Tables	Purpose
Dim_Date	date_key, full_date, day, month, quarter, year, day_name	Generated from rental_date, return_date, payment_date	Enables time-based trend analysis
Dim_Customer	customer_name, email, active_status, address, city, country	customer, address, city, country	Customer and location analysis
Dim_Film	title, release_year, language, rental_rate, rating, category	film, language, film_category, category	Film and category analysis
Dim_Store	store_address, city, country, manager_staff_id	store, address, city, country	Store performance and location analysis
Dim_Staff	staff_name, email, active_status, store_id	staff	Staff performance analysis
Dim_Inventory	inventory_id, film_id, store_id	inventory	Connects rentals to physical film copies

3.3.1 Dim_Date

Dim_Date is a generated dimension that does not come directly from the OLTP schema. Date keys are created from all unique dates found in rental_date, return_date, and payment_date columns. This dimension is shared by all three fact tables and enables time-based analysis at the day, month, quarter, and year levels.

3.3.2 Dim_Customer

Dim_Customer is built by joining the OLTP customer table with address, city, and country tables. Customer location attributes are denormalized into a single flat dimension to avoid joins during reporting. This supports both customer behavior analysis and location-based reporting.

3.3.3 Dim_Film

Dim_Film is built by joining the film table with language, film_category, and category tables. The category attribute is denormalized directly into the film dimension. This is a simplification of the many-to-many film-category relationship: since most films belong to a single primary category, embedding category into the film row is a practical design decision.

3.3.4 Dim_Store

Dim_Store is built by joining store with address, city, and country. Location data is denormalized for simpler querying. This dimension supports store performance and regional analysis.

3.3.5 Dim_Staff

Dim_Staff stores staff attributes including name, email, active status, and the store they belong to. This dimension supports staff performance analysis, allowing managers to compare rental and payment activity across employees.

3.3.6 Dim_Inventory

Dim_Inventory bridges the relationship between physical inventory items and their film and store associations. It is used in Fact_Rental and Fact_Inventory_Snapshot to connect rental transactions to specific copies of films held in specific stores.

3.4 Schema Type

The proposed schema is a star schema. Dimension tables are flat and denormalized, minimizing the number of joins required for typical analytical queries. Location data (city, country) is embedded inside Dim_Customer and Dim_Store rather than stored as separate normalized tables. This design choice prioritizes query simplicity and reporting performance over strict normalization.

4. Dimensional Model Diagram

The diagram below shows the proposed star schema for the Movie Rental Data Warehouse. The three fact tables (Fact_Rental, Fact_Payment, Fact_Inventory_Snapshot) are connected to their respective dimension tables through surrogate foreign keys. Dim_Date is shared across all three fact tables.

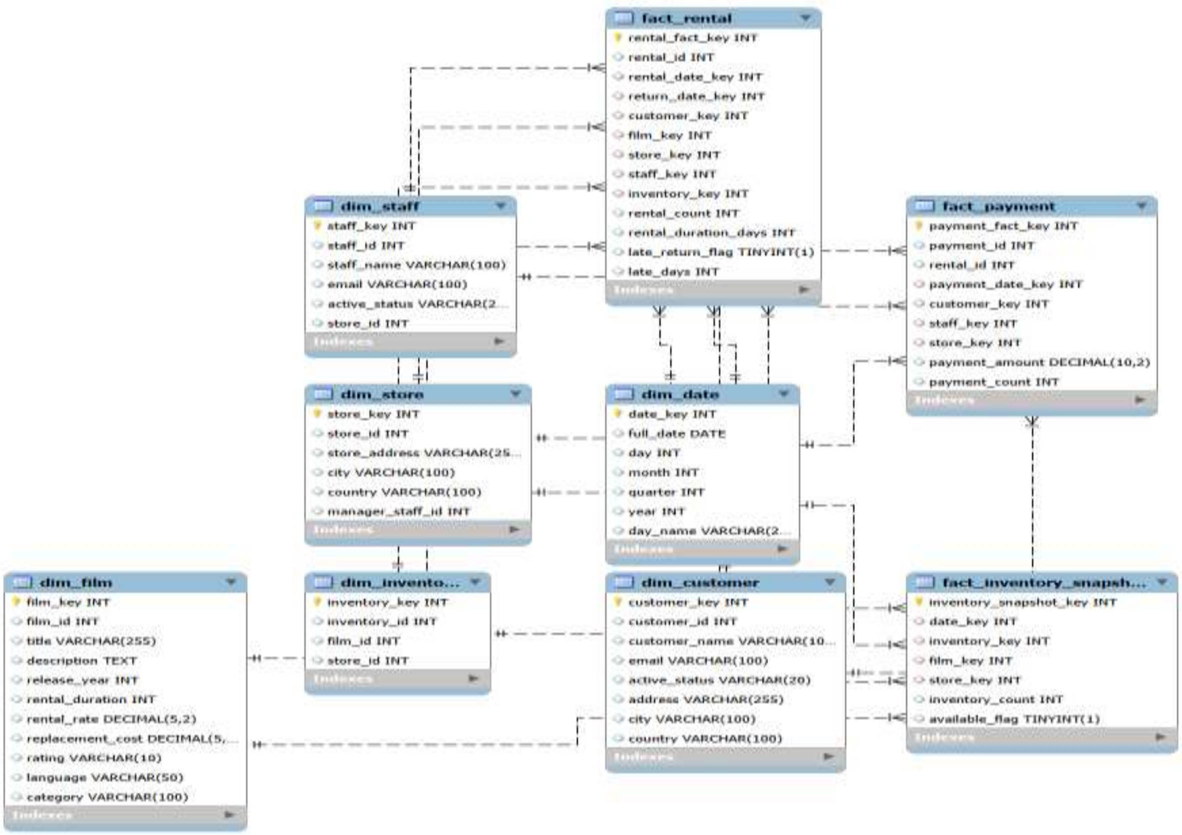

Figure 1: Star Schema Dimensional Model — Movie Rental Data Warehouse

Key relationships:

- Fact_Rental connects to: Dim_Date (rental date and return date), Dim_Customer, Dim_Film, Dim_Store, Dim_Staff, and Dim_Inventory
- Fact_Payment connects to: Dim_Date (payment date), Dim_Customer, Dim_Staff, and Dim_Store
- Fact_Inventory_Snapshot connects to: Dim_Date (snapshot date), Dim_Inventory, Dim_Film, and Dim_Store
- Dim_Date is a shared dimension used by all three fact tables
- Dim_Store and Dim_Customer are shared between Fact_Rental and Fact_Payment

5. ETL Design

The ETL (Extract, Transform, Load) process is responsible for moving data from the OLTP Sakila database into the data warehouse. The process follows three stages.

5.1 Extract

The following OLTP source tables are extracted and used to populate the data warehouse dimensions and fact tables:

Source Table	Used For	Reason
rental	Fact_Rental, Dim_Date	Core rental transaction records
payment	Fact_Payment, Dim_Date	Payment transaction records and revenue data
customer	Dim_Customer	Customer identifiers and attributes
film	Dim_Film	Film attributes including rental rate and duration
inventory	Dim_Inventory, Fact_Rental	Maps physical copies to films and stores
store	Dim_Store	Store identifiers and manager reference
staff	Dim_Staff	Staff identifiers, names, and store assignment
address	Dim_Customer, Dim_Store	Street address for customers and stores
city	Dim_Customer, Dim_Store	City names linked to addresses
country	Dim_Customer, Dim_Store	Country names linked to cities
category	Dim_Film	Film category names (Action, Comedy, etc.)
film_category	Dim_Film	Mapping table between films and categories
language	Dim_Film	Language of each film
actor	Not used directly	Could be used for actor-level analysis if needed
film_actor	Not used directly	Could support actor dimension if extended

5.2 Transform

Several transformations are required before data can be loaded into the data warehouse:

5.2.1 Joining and Denormalizing Dimension Data

- Join customer + address + city + country to build Dim_Customer as a flat table
- Join store + address + city + country to build Dim_Store as a flat table
- Join film + language + film_category + category to build Dim_Film with category embedded
- Combine first_name and last_name into a single customer_name or staff_name field (CONCAT)

5.2.2 Date Key Generation

- Extract all unique dates from rental_date, return_date, and payment_date columns
- Convert dates to integer date keys using format YYYYMMDD (e.g., 20050715)
- Populate Dim_Date with day, month, quarter, year, and day_name attributes
- Handle NULL return_date values by assigning a NULL or special 'Not Returned' date key

5.2.3 Surrogate Key Generation

- Generate auto-increment surrogate keys for all dimension tables (customer_key, film_key, store_key, staff_key, inventory_key)
- Map operational IDs from OLTP to surrogate keys during fact table loading

5.2.4 Measure Calculation

- Calculate rental_duration_days as DATEDIFF(return_date, rental_date)
- Calculate late_return_flag: 1 if rental_duration_days > film.rental_duration, else 0
- Calculate late_days: actual_duration - allowed_duration when late, else 0
- Use payment.amount directly as payment_amount in Fact_Payment

5.2.5 Data Cleaning

- Remove duplicate rental and payment records
- Standardize text fields: trim whitespace, standardize case for city, country, and category names
- Validate email format for customers and staff where available
- Handle open rentals (NULL return_date) by loading them with a NULL return_date_key
- Reject records with missing mandatory foreign key references

5.3 Load

Dimension tables must be loaded before fact tables because fact tables reference dimension surrogate keys. The recommended loading order is:

16. Dim_Date: load all date records first, as both Fact_Rental and Fact_Payment depend on it
17. Dim_Customer: load customer dimension using joined OLTP data
18. Dim_Film: load film dimension with language and category embedded
19. Dim_Store: load store dimension with address and location data
20. Dim_Staff: load staff dimension
21. Dim_Inventory: load inventory dimension mapping films to stores
22. Fact_Rental: load rental transactions using surrogate keys from all dimensions
23. Fact_Payment: load payment transactions referencing Dim_Date, Dim_Customer, Dim_Staff, Dim_Store
24. Fact_Inventory_Snapshot: load periodic inventory snapshot data last

For new dimension records (new customers, films, or stores), new rows are inserted with new surrogate keys. For changed records (e.g., a customer changes address), the design currently supports a Type 1 SCD approach (overwrite the existing record), which is appropriate for this high-level design. Future iterations may implement Type 2 SCD (add a new row with effective dates) if historical tracking of dimension changes is required.

6. Data Quality Considerations

The following data quality rules are applied during the ETL process to ensure the data warehouse contains accurate and consistent data:

Rule ID	Rule Description	Action on Failure
DQ-01	Every rental record must reference an existing customer_id in the source system	Reject record, log error
DQ-02	Every payment amount must be greater than or equal to zero	Reject negative amounts, log warning
DQ-03	rental_date must not be later than return_date when return_date is not NULL	Reject record, log error
DQ-04	Every film referenced in a rental must exist in Dim_Film	Reject record if film not found
DQ-05	Duplicate rental_id or payment_id values must not be loaded	Skip duplicate, log warning
DQ-06	Email values should follow a valid email format (contains @ and domain)	Load record, flag invalid email
DQ-07	City and country must not be NULL for customer and store dimensions	Load with 'Unknown' placeholder, flag
DQ-08	late_days must not be a negative value	Set to 0 if calculation yields negative
DQ-09	All fact table foreign keys must reference valid surrogate keys in dimension tables	Reject record, log referential integrity error
DQ-10	Missing return_date must be handled as open rental, not deleted	Load with NULL return_date_key

7. Sample Analytical Results

The following charts were generated from the data warehouse using Python and the Sakila dataset. They demonstrate the analytical capabilities of the proposed dimensional model.

7.1 Top Rented Films

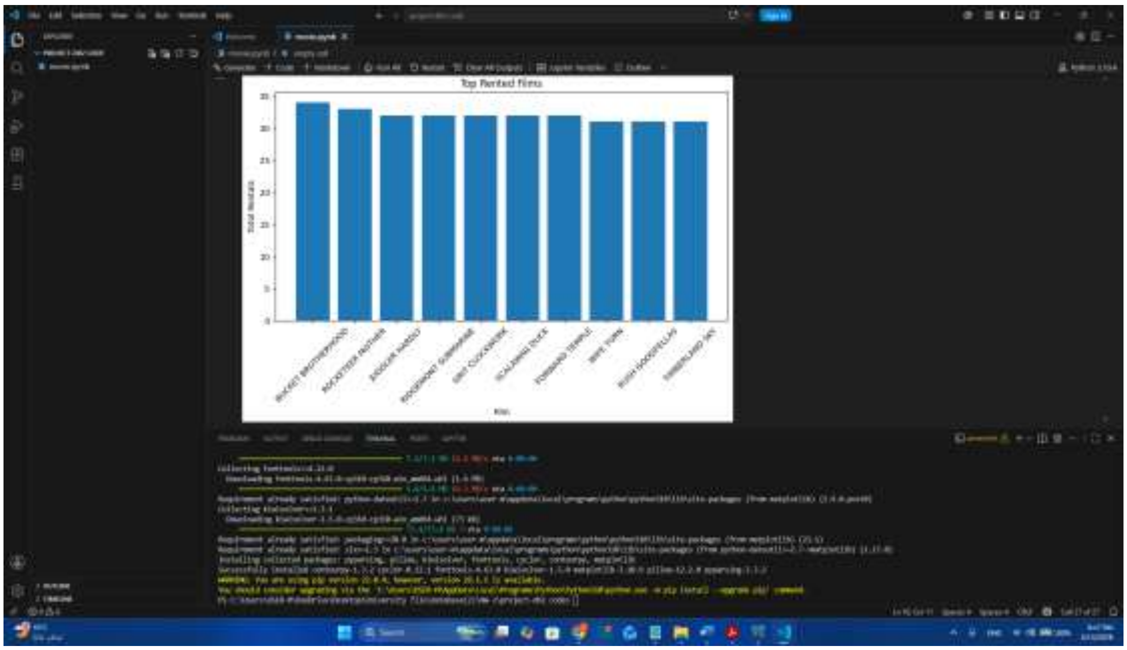

Figure 2: Top 10 Most Rented Films — Analysis from Fact_Rental joined with Dim_Film

The chart above shows the ten most frequently rented films. Bucket Brotherhood leads with approximately 34 rentals, followed closely by Rocketeer Mother and Juggler Hardly. This analysis is powered by aggregating rental_count in Fact_Rental grouped by film_key, joined to Dim_Film for film title.

7.2 Most Popular Film Categories

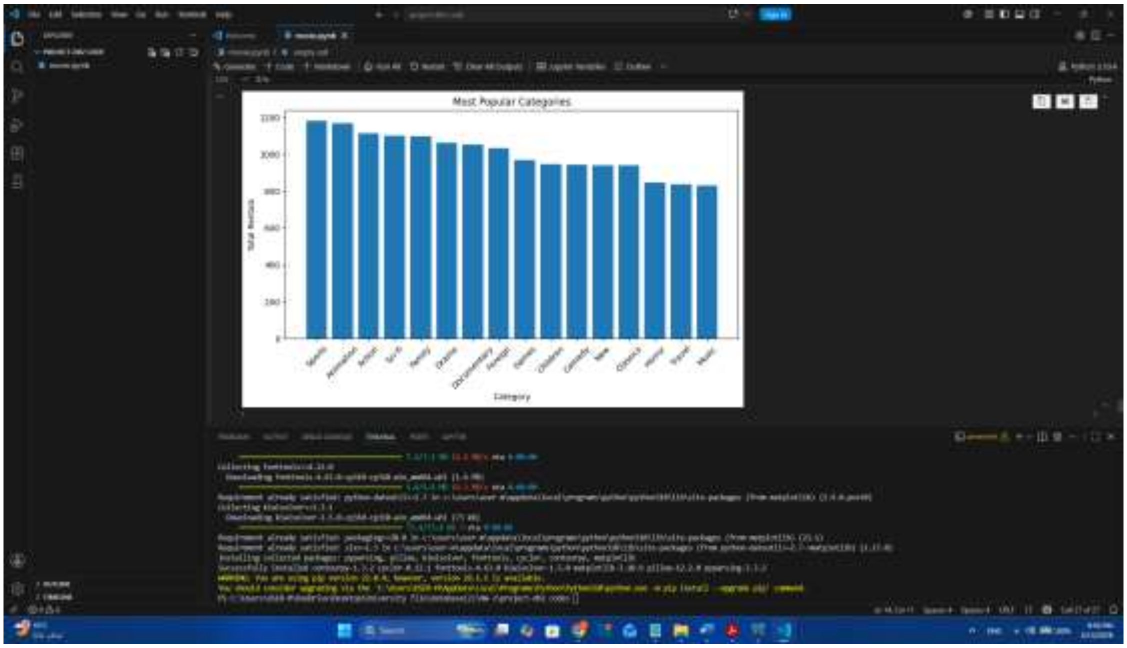

Figure 3: Most Popular Film Categories by Total Rentals — Analysis from Fact_Rental joined with Dim_Film

Sports and Animation are the top two film categories, each generating over 1,100 rentals. Action and Sci-Fi also perform strongly. This insight is generated by aggregating Fact_Rental records grouped by the category attribute stored in Dim_Film.

7.3 Revenue by Store

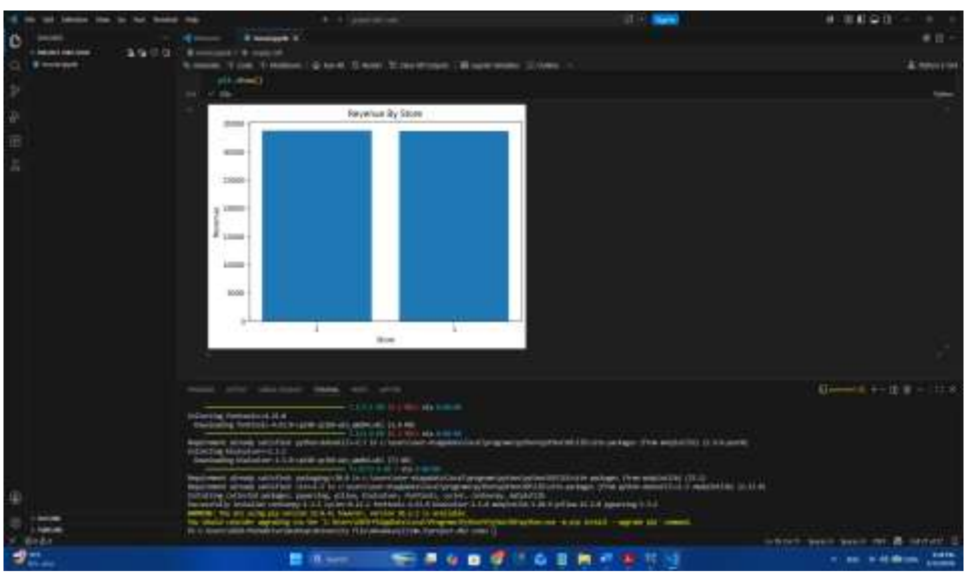

Figure 4: Total Revenue by Store — Analysis from Fact_Payment joined with Dim_Store

Both stores show similar revenue, each generating approximately 33,000 to 34,000 dollars in total. Store 2 shows slightly higher revenue than Store 1. This analysis is powered by summing payment_amount in Fact_Payment grouped by store_key, joined to Dim_Store.

7.4 Rental Activity Over Time

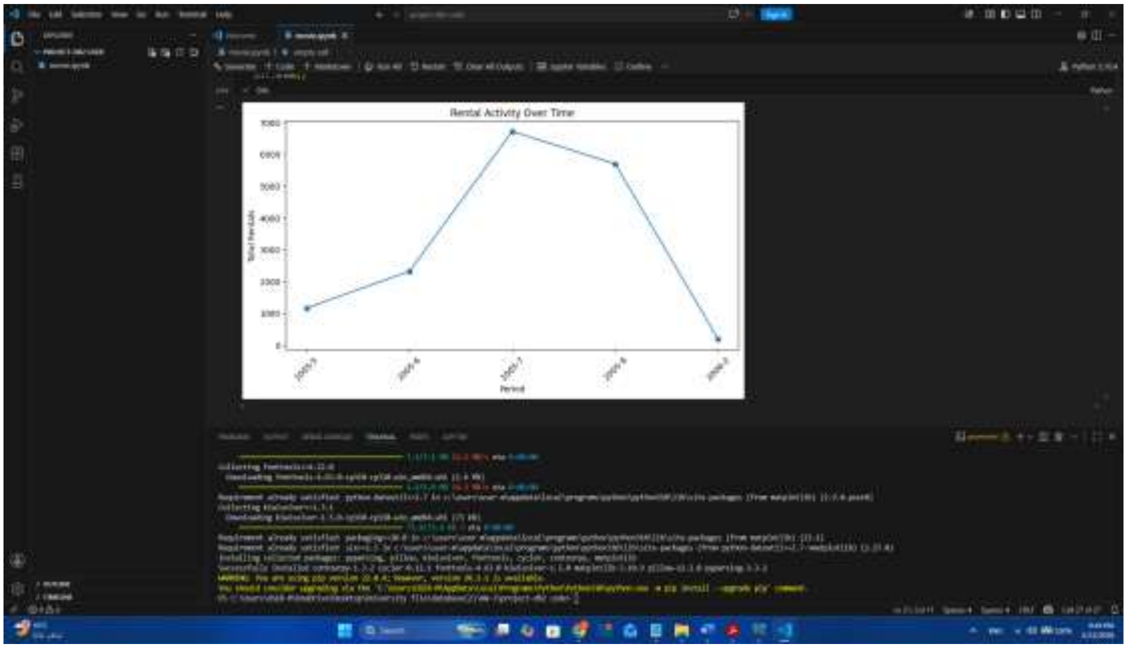

Figure 5: Rental Activity Over Time by Month — Analysis from Fact_Rental joined with Dim_Date

Rental activity shows a strong peak in July 2005 with approximately 6,700 rentals, followed by a decline in August 2005 to around 5,600 rentals, and a sharp drop to near zero in early 2006. This pattern suggests that the dataset covers a limited operational period. The time-based analysis is enabled by joining Fact_Rental with Dim_Date and grouping by year and month.

8. Conclusion

This report proposes a complete high-level data warehouse design for a movie rental business based on the Sakila OLTP database. The proposed solution reorganizes operational transactional data into a dimensional model that supports a wide range of analytical business questions.

The design includes three fact tables (Fact_Rental, Fact_Payment, and Fact_Inventory_Snapshot) and six-dimension tables (Dim_Date, Dim_Customer, Dim_Film, Dim_Store, Dim_Staff, and Dim_Inventory). The schema follows a star schema design, where dimension tables are flat and denormalized to minimize query complexity and improve analytical performance.

The ETL design covers extraction from 13 OLTP source tables, transformation steps including joins, surrogate key generation, date key creation, measure calculation, and data cleaning, and a structured loading order that ensures referential integrity. Ten data quality rules are defined to validate data before and after loading.

The sample analytical charts generated from the data warehouse demonstrate its practical value: the model successfully supports film popularity analysis, category performance analysis, store revenue comparison, and time-based rental trend analysis. These insights can directly support business decisions such as inventory procurement, staff evaluation, promotional campaigns, and store expansion planning.

Overall, the proposed data warehouse provides a solid analytical foundation that transforms operational movie rental data into actionable business intelligence.